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ABS CHALLENGES: AN ACCOUNTING INDEX OF WINS LEFT ON THE TABLE

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In 2026 Major League Baseball began letting teams challenge ball–strike calls against an automated ball–strike (ABS) system, and a team keeps every challenge that succeeds. How much do teams leave on the table by how they challenge? The published zone rule, applied to the public pitch locations, reproduces 99.95% of the 10,557 reviewed rulings, so the call the system would make can be reconstructed for unchallenged pitches too, and the challenge decision becomes a binary-response model fitted on challenged and unchallenged opportunities alike. The model identifies the total decision noise, not how much of it, or of the players' lapses, a new decision rule could remove. We define an accounting index, the wins per team-season by which a dynamic-programming benchmark learned from the data exceeds the teams' challenges when only the decision rule changes, estimate it out of fold (every learned component of the benchmark is fitted on other games and scored on held-out games), and treat the unidentified shares as sensitivity parameters. Under a model with player effects and lapses the index is 0.51 wins per team-season, with a bootstrap interval of [0.38, 0.63] that refits the whole procedure and is conditional on the model; it is 0.85 if the lapses could be removed, 1.38 if half the noise variance could, and 0.46–0.68 across the models that fit the decisions at least as well as a homogeneous one. Most of the gap is targeting: the rule "challenge if $p \geq 0.008/(g + 0.008)$", with p the probability that the call is wrong and g the win-probability stake, recovers 63% of it within the model at the teams' own challenge rate.

1. Introduction. From the 2026 season, Major League Baseball (MLB) calls balls and strikes with human umpires but lets the batter, catcher or pitcher appeal a call to the automated ball–strike (ABS) system, a Hawk-Eye measurement of the pitch against a rule-defined zone. Each team starts a game with two challenges; a challenge that overturns the call is kept and one that fails is lost. Over the regular season teams made 10,557 challenges, about two per team-game, of which 5,685 succeeded. Retention changes the economics: a challenge can be reused as often as it is right, so its value grows with the opportunities left and the right confidence threshold moves with the inning, the score and the inventory. MLB's Baseball Savant site prices a challenge at a fixed 0.2 runs (Baseball Savant, 2026a; Tango, 2026a); practitioners have proposed state-dependent heuristics (Hamm, 2026; Goldstein, 2026) and approximate dynamic programs (Arora, 2026; Namuduri, 2026); retained review rights were first modelled in tennis (Clarke and Norman, 2012; Nadimpalli and Hasenbein, 2013; Abramitzky et al., 2012).

The statistical problem. Valuing a challenge policy needs to know how well the challenger can tell that a call was wrong, and the data are decisions selected on exactly that judgement. Three difficulties follow, and this paper is organised around them. First, *identification of the response is not identification of the information*: even with the truth known for every pitch,

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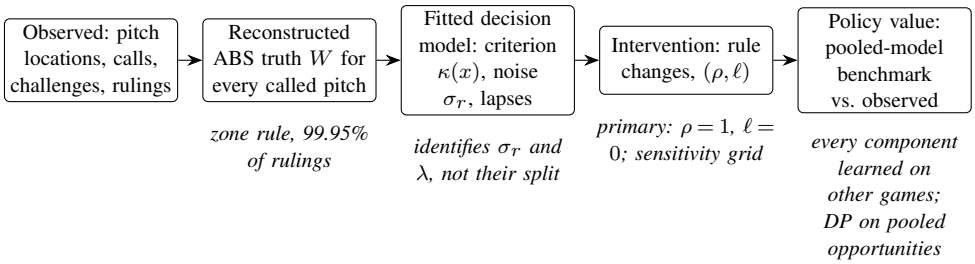


FIG 1. *From the observed data to the policy value. Each arrow adds an estimated object (above) and rests on an assumption or check (below).*

the challenge data identify the total noise between the truth and the decision, not how much of it is perception that no rule can remove and how much is a criterion that wobbles from pitch to pitch, or lapses of attention, that a disciplined rule could remove. What a team could gain therefore depends on an intervention the data cannot pin down. Second, *the benchmark is learned*: the stakes come from a win-probability engine fitted to the same season, and the decision model, the players' beliefs and the dynamic program are fitted to it too. A benchmark that selects pitches on estimated quantities and is scored on the same estimates is scored optimistically, as an estimated optimal policy is (Kitagawa and Tetenov, 2018; Athey and Wager, 2021). Third, *the decision model is uncertain*: the models that fit the decisions differ in what they imply, and the best-fitting one is the most expensive to refit.

Contributions. (i) A known truth: on the public 2026 coordinates the published zone rule reproduces the reviewed rulings to 99.95%, so the ABS label is reconstructed for unchallenged pitches as well, and the decision model is fitted on all eligible opportunities rather than on the selected challenges: the known truth removes the selective-labels problem that usually prevents evaluating human decisions against a rule (Lakkaraju et al., 2017; Kleinberg et al., 2018) (Section 3). (ii) An accounting index defined by an intervention: the primary intervention changes only the decision rule; the perceptual share ρ of the decision noise and the removable share ℓ of the lapses are not identified and are reported as a sensitivity grid. The index is an accounting quantity within a pooled decision model, not a causal effect and not a bound (Section 4). (iii) An out-of-fold estimator: a full policy cross-fit in which the engine, the decision model, the beliefs, the dynamic program and the rules are all learned on other games and applied unchanged to held-out games, whose flips and observed challenges are valued by an engine fitted to those games alone, in the spirit of cross-fitting in double machine learning (Chernozhukov et al., 2018); its target is the value of the benchmark a team could learn from data, not of an oracle, and its out-of-fold scoring removes the optimism of selecting on estimated stakes, as for estimated treatment rules (Hirano and Porter, 2009; Andrews, Kitagawa and McCloskey, 2024) (Section 6). (iv) Inference for the whole learned procedure under the best-fitting model: a bootstrap that refits everything, the decision model's hyperparameters included, in every replicate, and a coverage study on the real game stream (Section 7). Substantively, most of the gap is in which pitches teams challenge, not how often (Section 9). Figure 1 shows the chain from what is observed to the policy value, and which link carries which assumption.

2. The ABS challenge system and the data. *Rules.* After a called ball or strike the batter, catcher or pitcher may challenge immediately; the call is compared with the ABS zone, a rectangle whose width is the plate and whose top and bottom are fixed fractions of the batter's height, evaluated at the plate's midpoint. A successful challenge flips the call and the team keeps the challenge; a failed one is lost. A team that has used up its challenges

receives one in each extra inning (Major League Baseball, 2025). No challenge in the feeds is made with none left (Section S3.4 of the Supplementary Material; Janoušek and Pešta, 2026).

Data. We use the 2,425 ABS games of the 2026 regular season (2,429 completed, 4 at temporary venues without the system dropped). Pitch tracking comes from Statcast; the challenge events come from the MLB Stats API live feeds, whose counts reconcile with the league's totals and the feeds' own counters. A called pitch is joined to its Statcast record on its pre-pitch count and call type within the plate appearance: the count only rises, so the key is unique. Joining on the pitch number instead attaches the wrong location whenever one source counts an automatic ball or strike as a pitch and the other does not, which a check of the call description cannot detect when both pitches are balls or both strikes; 449 called pitches are numbered differently by the two sources, and on all of them the count-joined Statcast location agrees with the feed's own to at most 0.013 in. Only 3 called pitches find no count-consistent Statcast pitch; with those whose Statcast record has no coordinates, 201 called pitches are unlocated, and all 10,557 challenges are located. Of these, 4,766 were made by batters on called strikes (48.9% overturned), 5,612 by catchers on called balls (58.6%) and 179 by pitchers (39.7%). Pitches thrown by position players (1,143) are not challenge opportunities (Baseball Savant, 2026a). We pool pitchers with catchers as the fielding side.

Stakes. A count-level win-probability (WP) engine estimated from 2026 pitch transitions gives, for every called pitch, the stake g: the WP the challenging team gains if the call is flipped. Each count cell of the plate-appearance transition table is shrunk toward the table pooled over counts. In 331 called pitches this empirical engine makes a ball worth less than a strike to the batting team; a flip there is worth zero. The engine is league-average; Section 5 validates its contrasts.

3. A known truth and a decision model.

3.1. *The ABS truth.* Let d be the ball's radius plus the signed distance of the ball's centre to the ABS rectangle, in inches, positive when the ball touches the zone (the strike region is the rectangle grown by the radius, with rounded corners), computed from the Statcast location, and let e be the nearest edge (bottom, top, inside, outside). Write $w = \pm d$ with the sign chosen so that $w > 0$ means the umpire's call was wrong under the rule, and

$$W = \mathbf{1}\{w > 0\}, \qquad Y = \begin{cases} W & \text{with probability } 1 - \pi, \\ 1 - W & \text{with probability } \pi, \end{cases} \tag{1}$$

where Y is the ABS ruling on a challenged pitch and π a small reconstruction-disagreement rate (of the disagreements below, four lie within 0.005 in of an edge and one is a coordinate disagreement between sources), not a measured error rate of the ABS system. From 2026 Statcast reports the location at the plate's midpoint (Baseball Savant, 2026b); taken as published, these coordinates reproduce 99.95% of the 10,557 rulings under the rule (5 mismatches, all within 0.01 in of an edge or where Statcast and the live feed disagree on the location), square corners reproduce 99.71% (on an earlier join of 10,544 rulings), and shifting either zone line does not improve the fit. A smooth truth model with region offsets and Gaussian measurement noise is not preferred to the deterministic one, and BIC prefers (1) by 58; the fitted mislabel rate is $\pi = 0.05\%$. The agreement is measured on reviewed pitches, which cluster near the edges; on all called pitches the label from the live feed's own coordinates differs from the Statcast label for 0.02%, mostly within a quarter of an inch of an edge. Both sources share the Hawk-Eye measurement, so this checks the plane and the rounding, not the error against the ABS system's own measurement (Section S3). Statistical models of the human-called zone go back at least to Zimmerman, Tang and Huang (2019); here the zone is the rule's, and what we estimate is the challenger's perception of it.

3.2. *The decision model and its likelihood.* The player who may challenge (the batter on a called strike, the catcher on a called ball) observes a signal $s = w + \varepsilon$, $\varepsilon \sim N(0, \sigma_r^2)$, and challenges iff $s > \kappa(x)$, where x collects the challenges left, the inning, the count and $\log g$. In the main specification κ is piecewise linear in $\log g$ with hinges at fixed values (-5, -4, -3.2, about the 40th, 80th and 95th percentiles), and both κ and σ_r shift by zone region (Table S9 gives every term). For a called pitch at distance w,

$$\begin{aligned} \mathbb{P}(\text{challenge} \mid w) &= \Phi\left(\frac{w-\kappa}{\sigma_r}\right), \\ \mathbb{P}(\text{challenge}, Y = y \mid w) &= \Phi\left(\frac{w-\kappa}{\sigma_r}\right)\left[(1-\pi)\mathbf{1}\{y = W\} + \pi\,\mathbf{1}\{y \neq W\}\right]. \end{aligned} \tag{2}$$

The likelihood uses the challenge opportunities (a challenge held, the side that may challenge) within 8 in of the edge, 100,714 for batters and 141,199 for the fielding side; the window trims the fit only, and the policies see every opportunity. This is where the known truth does its work: w is observed for every called pitch, so the challenge curve is traced out over wrong calls that were not challenged as well as over those that were. A lapse rate λ, $\mathbb{P}(\text{challenge} \mid w) = (1-\lambda)\Phi\{(w-\kappa)/\sigma_r\}$ (Wichmann and Hill, 2001), captures the curve's levelling off below one far from the zone.

PROPOSITION 1. *Within a side and region, with κ fixed by the covariates: (a) the challenge curve (2) identifies (κ, σ_r) as soon as w takes two distinct values; (b) the log-likelihood is additive in π and in (κ, σ_r); (c) if the signed margin ABS measures equals the tracked margin plus independent $N(0, c^2)$ noise, the fitted σ_r is $(\sigma_r^2 + c^2)^{1/2}$ (an additive error in the scalar margin; Gaussian error in the two coordinates is not additive in the margin near a corner); (d) if the player sees $s_p = w + \sigma_p\varepsilon_1$ and challenges iff $s_p + \sigma_\nu\varepsilon_2 > \kappa(x)$, with independent standard normal $\varepsilon_1, \varepsilon_2$, the challenge curve depends on (σ_p, σ_ν) only through $\sigma_r^2 = \sigma_p^2 + \sigma_\nu^2$.*

Part (a) is the identification of a probit with an observed regressor (Jacobian determinant of absolute value $\phi_1\phi_2|w_1 - w_2|/\sigma_r^3$); with covariates it is a full-design condition, and at the fitted values the Jacobian of the challenge curve on a grid of distances has full column rank for both sides. Parts (a)–(d) are verified in Mathematica (Section S2). By (c), 0.3 in of tracking noise would bias the comparator's σ_r up by 0.5% for batters and 1.4% for catchers. Part (d) is the crux of this paper: the data identify the decision noise σ_r, not its perceptual part σ_p, and the same holds for the lapse rate, whose removable share is not identified. A lapse rate λ is identified only through the parametric shape: the response is still rising where the data end (73 batter and 47 catcher opportunities wrong by more than 4 in in the season), so λ is read from the curvature of the probit over $w \in (0, 4]$, not from an observed plateau (Prins, 2012). Simulating from the fitted model and refitting from a deliberately wrong start, the bias in σ_r over the 18 of 20 replicates whose refit converged is +0.4% for batters and +0.4% for catchers (Section S4).

3.3. *Estimates and the set of models.* Table 1 lists the decision models. The *primary model*, chosen before the game-state terms below were added, is player random effects on $(\kappa, \log\sigma_r)$, integrated with 11×11 Gauss–Hermite nodes, and a lapse rate. It is better than the homogeneous model by 1,295 BIC points. Its decision noise at the bottom edge is $\sigma_r = 2.62$ in for batters and 1.68 in for catchers, its lapse rates are 14% and 12%, and its player effects sit mostly in the criterion (SD 1.24 in for batters) rather than in the noise (SD of $\log\sigma_r$ 0.12). The fit converges to a maximum gradient below 10^{-6} with Newton steps on a finite-difference Hessian of the exact gradient. The *comparator* is the best homogeneous probit

TABLE 1

Decision models at least as good as the homogeneous comparator by BIC (all variants in Table S10). k: parameters; ΔBIC: relative to the comparator (negative is better); σ_r at the bottom edge. Gap: the accounting index under the primary intervention ($\rho = 1$, $\ell = 0$), wins per team-season, full policy cross-fit (Section 6), $R = 16$ worlds per split (Monte Carlo SE about 0.013), averaged over $S = 8$ splits for the primary model, the comparator and the comparator with state terms and $S = 4$ for the others; the split SD is 0.011–0.022, so the column resolves differences of about 0.02. In 2 of the 4 splits of the comparator with a lapse rate (2 of 4 with the time terms) a fold's fit stopped at its iteration limit.

Model	k	ΔBIC	σ_r batter / catcher (in)	Gap
Primary: player effects (11×11 nodes) + lapse	41	-1295	2.62 / 1.68	0.51
Primary + game-state terms in κ	51	-1527	2.62 / 1.63	0.53
Player effects, no lapse	39	-1283	2.81 / 1.81	0.68
Comparator (homogeneous): κ piecewise linear in $\log g$, κ, σ_r by region	33	$+0$	2.88 / 1.82	0.63
Comparator + game-state terms in κ	43	-220	2.87 / 1.80	0.65
Comparator + lapse rate	35	-5	2.68 / 1.69	0.46
Comparator, second-half shifts in κ and σ_r	39	-12	2.94 / 1.87	0.64
Same + lapse rate	41	-14	2.74 / 1.77	0.47

model, with $\sigma_r = 2.88$ in for batters and 1.82 in for catchers; among the pitches it would have challenged, 49.9% would have been overturned for batters against 48.8% observed, and 58.3% against 58.0% for the fielding side. Heavy-tailed laws are worse by 126–893 log-likelihood points at the comparator's parameter count. The hinge locations, the region structure and the comparator itself were chosen on these data, and no interval below accounts for that search. Three further homogeneous extensions fit better than the comparator and are carried through: a lapse rate (15% for batters, 12% for catchers), a criterion and noise that shift between March–June and July–September (BIC better by 12), and game-state terms in the criterion (the absolute score difference, late and close, two outs, a runner in scoring position, a runner on first; BIC better by 220). Added to the primary model the state terms improve BIC by another 232 and leave σ_r almost unchanged (2.62 and 1.63 in); since they move only the criterion, which the primary intervention replaces, we report that model as a row rather than switch the primary model.

4. The intervention, the benchmark and the estimand.

The intervention. A recommendation changes something, and what it can recover depends on what. The *primary intervention* changes only the decision rule: the player sees what they see now and attends as they attend now. By Proposition 1(d) the fitted noise may be partly criterion noise σ_ν that a disciplined rule removes; let $\rho = \sigma_p^2/\sigma_r^2$ be its perceptual share. In the lapse models let ℓ be the share of lapses a change of rule removes. The primary intervention is $\rho = 1$, $\ell = 0$: the benchmark player's signal is as noisy as the decisions reveal and every lapse stays. Secondary interventions take $\rho < 1$ (criterion noise removed) or $\ell > 0$ (lapses removed). The data cannot decide (ρ, ℓ), so Section 8 reports the loss on a grid, $\rho \in \{1, 0.75, 0.5, 0.25\}$ and $\ell \in \{0, 0.5, 1\}$; it does not reach the perfect-perception end $\rho \to 0$, where the gap is far larger. Within a model the benchmark's value is smallest at the primary intervention: the behavioural signal $s_p + \sigma_\nu \varepsilon_2$ is a garbling of s_p, so by Blackwell's theorem (Blackwell, 1953) the less noisy signal is worth at least as much to the same decision problem (checked in Mathematica on a grid of one-opportunity problems), and fewer lapses lose fewer opportunities. Blackwell's ordering is a statement about Bayes-optimal decisions;

for the estimated, pooled-model benchmark it is a regularity we check numerically (Table 5), and it orders benchmark values within the model, not what a team loses in real games.

The decision problem. Index the remaining opportunities of a team by t and the challenges held by c. At each opportunity a stake $g \geq 0$ and the player's calibrated probability p that the ruling will overturn the call arrive. A challenge gains g and keeps the challenge with probability p, and loses it otherwise. With $\Delta_t(c) = V_t(c) - V_t(c-1)$,

$$(3)\quad V_t(c) = V_{t-1}(c) + \mathbb{E}_t\big[(pg - (1-p)\Delta_{t-1}(c))^+\big], \qquad \text{challenge iff} \quad p \geq \frac{\Delta_{t-1}(c)}{g + \Delta_{t-1}(c)}.$$

The 0.2-run rule replaces Δ by a constant 0.2 runs and g by the run value of the flip; the "one-line rule" replaces Δ by a constant δ in WP.

ASSUMPTION 1. The pair (p, g) at opportunity t has law F_t, independent of the past and of c.

THEOREM 2. *Under Assumption 1, for all t and $c \geq 1$: $\Delta_t(c) \geq \Delta_t(c+1) \geq 0$ and $\Delta_t(c) \geq \Delta_{t-1}(c)$, so the optimal threshold falls as the game runs down and rises as challenges run out. For the rule "challenge iff $p \geq \delta/(g+\delta)$", with c_t the challenges it holds, $D = \Delta_{t-1}(c)$ and $I_t(c, g)$ the interval between $D/(g+D)$ and $\delta/(g+\delta)$,*

$$V_T^\star(c_0) - V_T^\delta(c_0) = \sum_t \sum_{c \geq 1} \mathbb{P}^\delta(c_t = c)\, \mathbb{E}_g \int_{I_t(c,g)} |p(g+D) - D|\, dF_t(p \mid g).$$

The DP we use (below) has the score in its state and a random number of opportunities, so it does not satisfy Assumption 1; the monotonicity in time (the marginal value of a challenge falling as the opportunities run out) fails at 20% of its computed entries and the concavity in inventory at 0.6% (Section S1), and the threshold movements we report are computed, not deduced from the theorem. The second part is the performance-difference identity (Kakade and Langford, 2002): the rule errs exactly on I_t and pays the Q-gap there. With a bounded density of p the integrand is second order in $\delta - D$, which is why the value curve can be flat; the calibrated posterior here is a step function, so that bound is only heuristic for the application, and the near-optimality of one constant below is an empirical finding (proofs are in Section S1, with stylised results on the square-root growth of a retained challenge and a flat-optimum bound that the calibrated game does not satisfy).

The benchmark. The calibrated p is the posterior $\mathbb{P}(Y = 1 \mid s, \text{cell})$ implied by the decision model, estimated by isotonic regression (Barlow et al., 1972) of simulated truth on the signal within cells of side, zone region, count class and stake tercile, on independent simulated draws. The DP is solved per team side; its state is the half-inning, the home score differential at the start of the half-inning (clipped at ± 6), the opportunity index and the challenges held. The number of a team's opportunities in a half-inning is random with the empirical law at that half-inning and score, every half-inning counting, including the 188 with no opportunity for either side; the stakes and beliefs of the opportunities are drawn from the pool of that half-inning and score; extra innings are an exact stationary block with the top-up rule. Outs, bases and count enter only through the stake, and a successful challenge changes neither the count process nor the runs that follow, so the DP is optimal for a pooled opportunity model, not for baseball. We call its policy the *pooled-model benchmark*. The loss against it is not a bound in either direction on the loss against the best policy for the real game process.

The estimand. For a policy a let $V(a)$ be the expected sum, over a team's season, of the WP stakes of the calls it flips, when the team follows a on a call stream drawn like the 2026 games, its opponents play as recorded, and the player's information is the signal, the count, the inning, the score, the challenges held and the stake. The benchmark is not known; it is learned from data, and we index the target by the learning. Let $\hat{a}_{-k}$ be the benchmark learned from a share $(K-1)/K$ of a season's games (engine, decision model, beliefs and DP), and let $V_k(\hat{a}_{-k})$ be its value on the remaining games. The *accounting index* is

$$\theta_K = \frac{1}{30}\,\mathbb{E}\Big[\sum_{k=1}^{K}\big\{V_k(\hat{a}_{-k}) - V_{\mathrm{obs},k}\big\}\Big], \tag{4}$$

wins per team-season under the primary intervention, with $V_{\mathrm{obs},k}$ the realised value of the teams' challenges on the games of fold k and the expectation over seasons of games like 2026, the benchmark's value being computed in the decision model's simulated worlds (Section 6). It is the gap to a benchmark a team could learn from most of a season, not to an oracle that knows the decision model, the beliefs and the stakes exactly; the oracle gap is larger by the cost of learning, which the coverage study of Section 7 puts at 0.01 for the comparator and 0.08 for the primary model. Since g is a jump in win probability, θ_K is in expected wins under the accounting that sums the jumps along the recorded game; we check that accounting, within the model, by re-simulating games (Section 7). It is an accounting index for one team that deviates alone, not an identified causal effect on real games; wins are zero-sum, so such gains would not add up across teams. A second reading compares the benchmark with the teams' fitted rule simulated in the same worlds rather than with their realised value (Section 7).

5. The stakes and their validation. Every value in this paper is a sum of ball-versus-strike contrasts g, so we validate the contrasts, not the levels of win probability (Section S5.8).

Statcast's own model. Statcast's win expectancy is an exact function of the pre-pitch state (no spread within any of its observed cells), so its ball-versus-strike contrast can be read off its own table for 94.5% of the called pitches. It correlates with ours at 0.978 (0.981 on challenged pitches); the mean of Statcast's contrast is 100.5% of ours, and 100.1% on the challenged pitches; the two agree in mean within every decile of leverage, though within a decile they rank pitches differently (correlations 0.17–0.96; Table S22).

Held-out outcomes. Within 2 in of the zone edge the call is decided by fractions of an inch of location; we *assume* that given the state, ball versus strike is then close to as good as random (a local-randomisation assumption; pre-pitch covariates are balanced between balls and strikes except the strike count, standardised difference 0.28, and we have not checked pitcher, umpire or pitch characteristics; the indicator is the call that stood, after any challenge). If the engine's contrast is right, the batting team's win indicator Y satisfies $\mathbb{E}(Y \mid x, \text{ball}) - \mathbb{E}(Y \mid x, \text{strike}) = g(x)$. Fitting the engine on a random half of the games and regressing $Y - \mathrm{WP}_{\mathrm{strike}}(x)$ on $\mathbf{1}\{\text{ball}\}\, g(x)$ in the other half (with $\mathrm{WP}_{\mathrm{strike}}$ as a control and game-clustered errors), the slope is 0.97 [0.71, 1.22] (78,123 pitches; mean over five splits), 0.90 [0.63, 1.17] in the top leverage tercile, where the gains are, 0.97 for Statcast's contrast, and 0.84 [0.59, 1.09] with the umpire's original call. A slope of one is what a correct contrast predicts; the intervals are wide, so this is a consistency check that cannot detect an error of about a quarter in the contrasts, not a validation to that precision. A common rescaling of g rescales the benchmark's continuation values and the gap by the same factor and leaves its decisions unchanged; a fixed constant δ would have to be rescaled too.

A second yardstick. Scoring every flip at Statcast's independent contrast instead, with decisions on our full-season engine, gives 0.72 against 0.74 on the same pitches (comparator,

in-sample); re-running the comparator's in-sample analysis (refit, not cross-fitted) on Statcast's stakes, on the shortened stream of the 94.5% of called pitches that have a Statcast contrast, gives 0.75 and the same ranking of every rule.

6. Out-of-fold evaluation: the full policy cross-fit. Every learned object is a source of optimism when the benchmark is scored on the games it was learned from: a policy that selects pitches on estimated stakes picks those whose stake is overestimated; beliefs and continuation values tuned to the evaluation games fit their noise; and a decision model fitted to them shapes the worlds the benchmark is scored in. Cross-fitting the whole policy removes the first two and the third's influence on the benchmark's decisions; it does not remove the decision model's error from the simulated worlds in which those decisions are scored, which is why the model range and the coverage under misspecification are reported beside the interval. The games are split at random into $K = 5$ folds (inside a bootstrap resample, over the original games, so that copies of a game share a fold). For fold k:

1. *Learned on the other folds*: the WP engine and the stakes every decision uses, for the held-out pitches too; the DP's run laws; the decision model, by maximum likelihood on the training opportunities with their training stakes, and for the primary model the players' posterior effects; the stake terciles of the belief cells and the belief maps, by isotonic regression of simulated truths on simulated signals of the training pitches; the DP's continuation values; and the one-line rule's best constant.
2. *Applied, unchanged, to the held-out games*: truths and signals are drawn from the training fit, the player's belief is the training map of the pitch's cell applied to its signal, and the frozen DP and rules decide along the recorded stream, holding the challenges they would hold.
3. *Scored on the held-out games alone*: every successful flip, of the benchmark, of each rule and of the teams' observed challenges, is valued at the signed contrast of an engine fitted to fold k alone (signed, since clipping a noisy contrast at zero biases it up).

The K folds cover the season once, so the sum over folds estimates (4); $\hat{\theta}_K$ averages $S = 8$ random splits with $R = 16$ simulated worlds in each.

REMARK 1 (Out-of-fold scoring). No decision uses a quantity estimated from fold k's plate-appearance outcomes: the decision stakes, the decision model, the beliefs and the DP are fitted on the other folds. A policy therefore cannot select pitches on the scoring engine's estimation error, which is where in-sample scoring is optimistic: for a Gaussian estimate with SD s and a cutoff c, selecting on the estimate and scoring at it overstates the value by $s\,\phi\{(c-g)/s\}$, and scoring at an independent estimate removes the excess (Section S2). This is not a full independence argument. The decisions still depend on fold k through the states its games visit, and those states are part of the transitions the fold engine is fitted on, so conditioning on them would nearly fix the scoring contrast; and the observed challenges, real decisions in the held-out games, are outside the argument altogether. We check the residual dependence numerically instead: the fold engine's contrast is unbiased against the full-season engine's on the same pitches, with a ratio of sums of 1.001 over all called pitches, 1.000 over the challenged ones and 0.993–1.004 within deciles of the stake (src/scoring_bias.py).

One input is not split: the noise quintiles that define the primary model's belief cells are cut over all players' fitted noise, a covariate rather than an outcome; cutting them on the training rows only gives 0.51. The same split makes the benchmark's beliefs and DP out-of-sample objects, so (4) is the value of a benchmark a team could learn, which is what an assessment of the teams' play should compare them with.

TABLE 2

The gap, wins per team-season, by estimator and reading, primary intervention unless stated. In sample: $R = 40$ worlds; stakes cross-fitted only: $S = 8$ (comparator) and $S = 4$ (primary) splits; full policy cross-fit: $S = 8$, $R = 16$ (Monte Carlo SE about 0.013 within a split). The last row is from the coverage study (Section 7). Every later number in the paper is one of these cells or is labelled otherwise.

Estimator or reading	Comparator	Primary
In sample: everything fitted and scored on all games	0.70	0.59
Stakes cross-fitted only	0.65	0.55
Full policy cross-fit (the index)	0.63	0.51
against the teams' fitted rule, simulated	0.68	0.59
every lapse removable ($\ell = 1$)	—	0.85
half the noise variance removable ($\rho = 0.5$)	1.62	1.38
Oracle minus learned benchmark (coverage study)	0.01	0.08

What each layer costs. Table 2 collects every estimator and reading of the gap. For the comparator, scoring in-sample at the full-season stakes gives 0.70; cross-fitting only the stakes, with the decision model, beliefs and DP fitted on all games, gives 0.65; the full policy cross-fit gives 0.63. The selection on estimated stakes is most of the optimism; learning the rest out of sample costs a few hundredths.

The split is between random folds of one season, not over time: a constant chosen on March–June and frozen with every other component of the policy, including a decision model fitted on the stakes of a March–June engine, keeps 95.6% of the frozen benchmark on July–September.

7. Policy values.

Design. Every policy is evaluated on the same simulated worlds: every real 2026 called pitch keeps its known truth, the ruling is drawn with the mislabel rate, the signal is drawn from the fitted model, the belief is computed from it, and the policy decides along the real call stream at every opportunity (369,619 pitches), holding the challenges it would hold. For the primary model each world also draws the players' effects jointly from their posteriors. All values are full policy cross-fits (Section 6).

Inference. The interval comes from a bootstrap over games (Efron and Tibshirani, 1993) that repeats the whole procedure in every replicate ($B = 400$ for each model): the games are resampled with their Statcast records, a fresh split into folds is drawn over the original games, and within each fold the engine, the run laws, the decision model — for the primary model all its hyperparameters, refitted from a warm start, and the players' posteriors — the beliefs, the DP and the rules are learned again and scored on the held-out games ($R = 8$ worlds; a replicate of the primary model takes about 137 minutes on one worker). Resampling games targets a population of games exchangeable with the 2026 games (rosters and schedule as in 2026). We report the normal interval, the point ± 1.96 bootstrap SDs: [0.38, 0.63] for the primary model and [0.51, 0.76] for the comparator (percentile [0.40, 0.66] and [0.48, 0.73], basic [0.35, 0.61] and [0.53, 0.79]; bootstrap means 0.53 and 0.61). The bootstrap mean differs from the point, upwards for the primary model and downwards for the comparator, because a resample repeats about a third of its games: each training fold then learns the engine, the beliefs and the DP from fewer distinct games, and the players' posteriors from duplicated data. The bootstrap's spread is right while its location is not: in the coverage study the bootstrap SD matches the spread of the point across simulated seasons (0.063 against 0.061 for the primary model), and the intervals that use the bootstrap's location cover less. Hence the normal interval. The choice was made after the coverage study; with 30 seasons the coverage

rates carry a Monte Carlo error of several percentage points, so the study ranks the intervals but does not calibrate them finely. A replicate also uses one random split and $R = 8$ worlds where the point averages $S = 8$ splits of $R = 16$; that algorithmic noise is 12% of the primary bootstrap variance (without it the SD would be 0.060 rather than 0.064), so the intervals are slightly conservative on this account. The two models use different convergence criteria: the primary model's Newton refinement stops at a maximum gradient below 10^{-2}, which all 400 replicates reached; the comparator's trust-region fit stops at 10^{-4} within 60 iterations, which 22 replicates missed in some fold and which are kept (Section S5.11). The primary model's likelihood is integrated with 11×11 nodes; at its fitted parameters 31×31 nodes change the negative log-likelihood by 3.2 points (src/quadrature_check.py), and we have not refitted with more nodes. The interval is not a guarantee for a non-smooth functional: the benchmark is an argmax over thresholds and the beliefs are isotonic fits, the setting in which bootstraps of estimated optimal policy values can fail (Chakraborty, Laber and Zhao, 2013; Luedtke and van der Laan, 2016). The index sums thousands of threshold decisions, which smooths it, but we rely on the coverage study, not on a theorem. The intervals are conditional on the decision model and the intervention: they carry sampling, learning and engine uncertainty, not the model class or (ρ, ℓ), which Section 8 treats as a sensitivity analysis.

Coverage. We check the intervals on simulated seasons that keep the real 2026 game stream, so that no game repeats as it would in a resampled season, and simulate the challenges along it from a generating model; the analysis is the whole procedure above. Two targets are computed under each generating model: the cross-fit's own target θ_K of (4), the benchmark learned from a simulated season and valued in the generating model's world, and the oracle gap of a benchmark that knows the generating model. With the primary model generating and analysing (30 seasons, $B = 30$), the point is off θ_K by 0.009 (SE 0.011), its spread across seasons (0.061) matches the mean bootstrap SD (0.063), and the normal interval covers θ_K in 100% of seasons, against 80% for the percentile and 80% for the basic interval. The oracle gap is larger by the cost of learning (0.08) and is covered by the normal interval in 77%. With the comparator analysing its own data (50 seasons, $B = 50$) the point is off by -0.032 and the normal interval covers in 90%. A model without lapses or player effects, applied to data that have them, shifts the interval rather than widening it: the comparator analysing data from the lapse model and from the primary model (30 and 30 seasons) covers θ_K in 10% and 27% and overstates it by 0.212 and 0.160; the primary model analysing data from player effects *without* lapses, where it estimates lapses that are not there, stays on target: it covers in 100% of 12 seasons and is off by -0.001 (Section S5.11). The richer model protects against the simpler world; the converse does not hold. The seasons keep the 2026 stream and its plate-appearance outcomes, so the study checks the behavioural component of the interval, not the game-resampling component.

Policies. Besides the benchmark we run, in the same cross-fit, the one-line rule with the constant $\delta = 0.008$ of our earlier analysis, with $\delta = 0.01$ and with the constant that does best on the training folds, the fixed 0.2-run rule of Baseball Savant (Baseball Savant, 2026a), "challenge iff $p \geq 0.2/(0.2 + \mathrm{RV})$" with RV the run value of the flip, and the teams' fitted behavioural rule $s > \kappa(x)$. Under a lapse model every rule is played by the benchmark's player, who lapses on the same pitches. All rules use the same calibrated beliefs, which nobody using them in 2026 had, so this compares the rules' targeting, not their real-world use. We also run the other public rules — the runs-objective rule of Namuduri (2026), the 48-cell card of Arora (2026) and Savant's form with a tuned run cost — and the teams' own rule re-tuned, in the same cross-fit with every tuned object learned on the training folds (Table 4; 8 splits).

TABLE 3

Benchmark and observed play, wins per team-season (sum of WP increments), full policy cross-fit, primary intervention, with 95% intervals from the bootstrap that refits the whole procedure ($B = 400$ for each model): percentile for the wins, normal for the index, whose bootstrap location is shifted (Section 7). Challenges per team-game. The observed value differs in the third decimal between the models because the scoring engines depend on the random splits.

	Primary		Comparator	
Policy	Wins	Challenges	Wins	Challenges
Benchmark (primary intervention)	3.85 [3.71, 4.06]	2.71	3.98 [3.79, 4.11]	2.70
Teams' fitted rule $s > \kappa(x)$, simulated	3.26	2.16	3.30	2.16
Observed 2026 (realised)	3.34 [3.17, 3.51]	2.18	3.34 [3.17, 3.51]	2.18
Accounting index	0.51 [0.38, 0.63]		0.63 [0.51, 0.76]	

TABLE 4

Rules a team could implement, on the same worlds as Table 3: the share of the gap between the observed challenges and the benchmark that the rule recovers, in percent (95% bootstrap percentile intervals where the rule was in the bootstrap), and challenges per team-game (Chall.). A negative share means the rule earns less than the teams did. Every tuned object (constants, run cost, card thresholds, re-tuned criteria) is learned on the training folds.

	Primary		Comparator	
Rule	Gap (%)	Chall.	Gap (%)	Chall.
One-line rule, $\delta = 0.008$	63 [54, 74]	2.17	70 [61, 76]	2.22
One-line rule, δ chosen on the training folds	64 [54, 73]	2.27	69 [59, 75]	2.25
One-line rule, $\delta = 0.01$	43	1.88	56	1.90
Runs-objective rule (Namuduri, 2026)	87	2.96	87	2.97
48-cell card, published (Arora, 2026)	60	2.54	65	2.59
48-cell card, training-DP thresholds	56	2.55	58	2.51
Savant's form, run cost tuned on training folds	42	2.62	50	2.55
Fixed 0.2-run rule (Baseball Savant, 2026a)	−172 [−219, −99]	1.12	−142 [−209, −109]	1.03
Teams' rule, intercept re-tuned	40	2.94	50	3.03
Teams' rule, slope on $\log g$ re-tuned	63	2.38	79	2.45
Teams' rule, intercept and slope re-tuned	64	2.42	79	2.47
Teams' fitted rule $s > \kappa(x)$, simulated	−16	2.16	−7	2.16
Observed 2026	0	2.18	0	2.18
Benchmark	100	2.71	100	2.70

Results. Under the primary model the benchmark gains 3.85 [3.71, 4.06] wins per team-season from challenges at 2.71 challenges per team-game; teams realised 3.34 [3.17, 3.51] at 2.18. The accounting index is 0.51 [0.38, 0.63] wins per team-season. Under the comparator

it is 0.63 [0.51, 0.76], and across the 8 models at least as good as the comparator by BIC it ranges over 0.46–0.68 (Table 1; a sensitivity range, not a confidence interval). The models with a lapse rate give the smaller values: a benchmark played by a player who still lapses has fewer opportunities to exploit. The teams' fitted rule, simulated in the same worlds, earns 3.26 against the 3.34 teams realised; as a share of the gap, -16% [-45, 0], so under the primary model the simulated behaviour falls detectably short of the real one in value, while under the comparator it does not (-7% [-30, 6]). Against the simulated rule instead of realised play the gap is 0.59 [0.54, 0.70] (comparator 0.68): the second reading of Section 4, with the benchmark and the behaviour in the same model worlds. Under the comparator with the stakes cross-fitted only, batters account for 0.38 [0.30, 0.47] and catchers and pitchers for 0.27 [0.18, 0.35], a split of who makes the flips rather than a causal attribution, since the two sides share one inventory.

The value curve is flat near its optimum in share of the benchmark, and steeper in share of the gap. The one-line rule with $\delta = 0.008$, the best constant on the full 2026 season in our earlier, in-sample analysis, gets 95.2% of the benchmark and recovers 63% [54, 74] of the gap under the primary model (70% under the comparator); with the constant chosen on the training folds (averaging 0.0069–0.0077 across folds) it recovers 64%; the best constant on the grid is 0.007, with 95.5% of the benchmark, while $\delta = 0.01$ recovers only 43% of the gap. At the teams' own challenge rate the one-line rule recovers 63% [53, 72]. In the same cross-fit, with every tuned object learned on the training folds, the runs-objective rule of Namuduri (2026) recovers 87% of the gap, more than the one-line rule, but it needs a run-value DP solved for every state, while the one-line rule needs only a table of WP stakes; the 48-cell card of Arora (2026) recovers 60% with its published thresholds and 56% with each cell's median break-even from the training DP. The fixed 0.2-run rule, which Baseball Savant publishes as the break-even for a challenge, gets 64.3% of the benchmark, less than the teams' 86.9%, and challenges far too rarely (1.12 per team-game); with its run cost tuned on the training folds the same form recovers 42% of the gap. This is a verdict on the constant, which Savant publishes as the break-even of a single challenge, with beliefs that nobody using it had, not on its users.

Causal accounting. Summing stakes along the recorded game is accounting: after a flipped call the rest of the game would have differed. We re-simulate in a pitch-level simulator (events, in-play outcomes and baserunning by count, outs and bases from 2026 Statcast; criterion and belief recomputed at every simulated state), comparing like with like. The re-simulation uses the comparator with the full-season stakes, so it compares with the uncorrected 0.70 rather than the cross-fitted index. *Paired*: the benchmark and the teams' simulated behavioural rule are played along the stream in the same world, and from the first pitch at which their flips differ both are re-simulated to the end of the game, each with its own inventory; the benchmark's gain is 0.71 [0.67, 0.76] wins per team-season valued by later stakes and 0.68 counting simulated wins, against 0.72 by the path sum on the same worlds. *From the recorded game*: branching from the first pitch at which the benchmark and the recorded challenges differ, and continuing the team with its fitted rule, gives 0.72 against 0.69 for the path sum against the observed challenges. The win count sits about 0.03 below the stake-valued estimate; part of that is the simulator's flip swing, 1.2% short of g, and the rest we cannot attribute (Section S5.4). The check is internal to the model: the engine and the simulator share their structure.

8. What the data cannot decide. The index is defined for the primary intervention, which changes only the decision rule. Table 5 reports it, with the same full policy cross-fit, over the perceptual share ρ of the decision noise and the removable share ℓ of the lapses, for the primary model and the comparator (the grid for the other models, cross-fitting the stakes

TABLE 5

The accounting index, wins per team-season, full policy cross-fit ($S = 8$ splits), by intervention: ρ is the perceptual share of the decision noise (1: all of it stays), ℓ the share of lapses a rule change removes (primary model). Primary intervention: $\rho = 1$, $\ell = 0$. A sensitivity grid, not a confidence set.

Model (ℓ)	$\rho = 1$	$\rho = 0.75$	$\rho = 0.5$	$\rho = 0.25$
Primary ($\ell = 0$)	0.51	0.86	1.38	2.30
Primary ($\ell = 0.5$)	0.68	1.06	1.62	2.62
Primary ($\ell = 1$)	0.85	1.25	1.86	2.93
Comparator	0.63	1.02	1.62	2.68

only, is in Section S5.9). The grid is a structural sensitivity analysis within the Gaussian decision model, not a sharp identified set (Manski, 2003): we do not claim that the data bound (ρ, ℓ), and we do not attach inference to the grid's corners. Moving away from the primary intervention raises the index, and steeply. Removing all lapses raises it from 0.51 to 0.85 under the primary model; removing half the variance of the decision noise ($\rho = 0.5$, about 29% of its SD) raises it to 1.38 (1.62 under the comparator), and the largest value on the grid, which stops at $\rho = 0.25$, is 2.93. The value of better information is large because the benchmark exploits sharper beliefs on every pitch. Within each model the primary intervention gives the smallest value on the grid, as Blackwell's ordering predicts for a Bayes-optimal decision maker (Blackwell, 1953); for the estimated benchmark this is a regularity of the table, not a theorem. The lapse corner matters as much as the noise: the far misses that are not challenged are partly a choice, since their challenge rate rises with the stake and falls with one challenge left (Section S4), so some of the lapses may be removable. This orders benchmark values within the pooled model; it is not a lower bound on what a team loses in real games. How much more a team could gain by changing how its players see or attend the data cannot settle; only an intervention or an independent measurement of perception could.

9. The mechanism: targeting, not frequency. The benchmark challenges 24% more often than teams, but frequency is not where most of the gap is. Under the primary model the one-line rule challenges 2.17 times per team-game against the teams' 2.18 and recovers 63% of the gap; at exactly the teams' rate (a mixture of the two grid constants that bracket it) it recovers 63% [53, 72]. The teams' own rule class, which needs no calibrated belief, says the same. In the full policy cross-fit we re-tune the criterion $s > \kappa(x)$ of each side on the training folds and apply it, frozen, to the held-out games: shifting only its intercept recovers 40% of the gap, changing only its slope on $\log g$ 63%, and both 64% (comparator: 50%, 79% and 79%). A team that teaches its players to demand less evidence when more is at stake, without changing how often they challenge on average, recovers most of what the benchmark gains. These are comparisons of interventions within a rule class, not an identified decomposition: changing a criterion's slope also changes its frequency, timing and inventory. The threshold slopes below are for the comparator.

The threshold slopes say the same. Mapping the fitted criterion to the confidence threshold p^* it implies and regressing $\log\{(1 - p^*)/p^*\}$ on $\log g$ with the criterion's state controls, teams' slope is 0.08 [0.06, 0.10] for batters and 0.04 [0.01, 0.07] for catchers; the benchmark's own thresholds in the same regression on the same pitches give 0.58 [0.55, 0.61] and 0.59 [0.57, 0.62] (normal intervals from the bootstrap SD); with the twelve exact count cells as controls, 0.08 and 0.04 against 0.56 and 0.58. On calls wrong by less than an inch, batters challenge 12% in the lowest stake quintile and 29% in the highest. The stake is the engine's g; players who respond to a cruder perceived stake may respond to it more strongly, so this concerns the engine's stake, not what players attend to. The game-state terms of Section 3.3

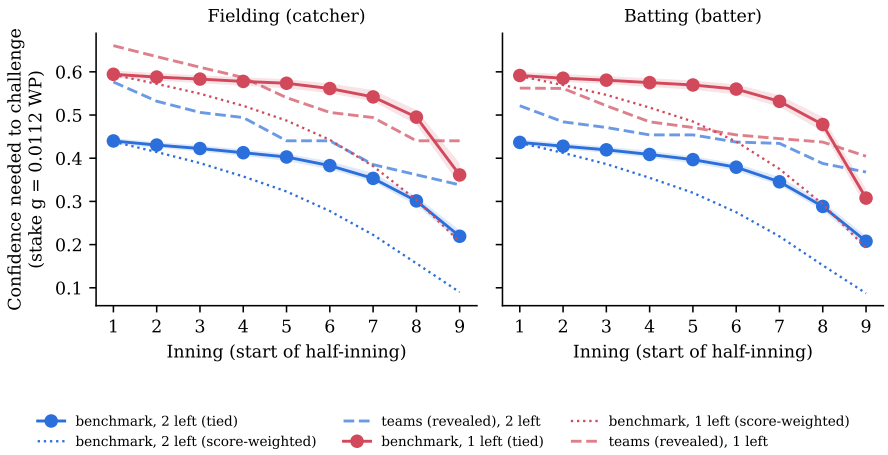


FIG 2. *Benchmark confidence threshold for a challenge at the reference pitch (median stake of the challenged pitches), by inning and challenges held: tied score with a 95% band over refits of the comparator on game resamples (engine held fixed, so the band is narrow), average over the score states visited in 2026, and teams' revealed thresholds. Panels by challenging role.*

point the same way: the criterion falls with two outs (-0.32 in for batters, -0.38 in for catchers) and with a runner in scoring position, so players do respond to leverage that the engine's g also carries, and part of the low slope on $\log g$ is that response routed through the state terms.

Figure 2 shows how the benchmark's threshold moves. Inventory moves it at every score: in the fifth inning, holding one challenge rather than two raises it by 17.0 [16.6, 17.4] percentage points in a tied game. Time moves it mainly through the score: averaged over the score states games visit, the threshold with one challenge left falls by 27.5 [26.6, 28.7] points from the first to the eighth inning, but in a game that stays tied by only 8.5 [7.1, 9.8], because a late tie keeps every remaining call valuable.

10. Robustness. *Over the season.* Fitted to March–June and predicting July–September (Table 6), batters' decision noise is stable (2.88 and 2.87 in) but the fielding side's falls from 1.92 to 1.71 in, and catchers challenged 22% more than the first-half fit predicts. The time-varying model of Table 1 absorbs this drift and gives 0.64 under the primary intervention; by calendar month the comparator's gap ranges from 0.41 to 0.92.

Game state in the criterion. With the state terms the index is 0.53 under the primary model and 0.65 under the comparator, against 0.51 and 0.63 without them.

The estimator's own choices. How the benchmark's beliefs are learned matters. With four times as many simulated draws for the belief maps the primary index is 0.53 instead of 0.51; with the beliefs calibrated by the exact count rather than three count classes it is 0.44 (0.39 with the usual draws; comparator 0.60), and few held-out pitches fall in cells unseen in training, so this is not a fallback effect. The calibration cells are a modelling choice about what a player could learn, and the index moves by about 0.1 with it; with $K = 2$ and $K = 10$ folds the comparator gives 0.61 and 0.67 against 0.63 at $K = 5$, the cost of learning from a smaller share of the season.

Recovery. Simulating the season from the primary model and refitting it, the lapse rates are recovered with bias +0.9 and -1.0 percentage points (SD across replicates 3.6 and 2.4) and σ_r with -1.2% and +0.0% over 12 replicates: under the model the lapse is identified, by the shape, though not precisely. The player-effect SDs come back smaller (-17% for the batters' criterion), because the simulated players sit at their shrunken posterior means.

TABLE 6

Out-of-sample and external-league checks of the homogeneous comparator. Overturn rate: among the pitches the model would have challenged, against the observed rate among the challenges made.

Fit sample → evaluation	σ_r (in)		Overturn rate, model / obs. (%)	
	batter	catcher	batter	catcher
MLB 2026 Mar–Jun → Jul–Sep	2.88	1.92	50.4 / 50.2	58.3 / 57.7
Triple-A 2025 (refit)	3.22	2.09	46.3 / 45.0	54.6 / 53.8
Spring training 2026 (refit)	3.78	2.18	46.3 / 45.7	57.5 / 59.5

Other leagues and coordinates. Fits of the comparator to Triple-A 2025 and spring training 2026, whose feeds carry the ABS system's own measurements, give decision noise 12–15% and 20–31% above MLB's. On the MLB feed's own coordinates the rule reproduces 99.97% of the rulings and σ_r is 2.90 and 1.84 in.

Model checks. By side and zone region, the isotonic regression of the challenge indicator on w stays within 8 percentage points of the fitted probit curve (pointwise graphical checks, Figure S1). Logistic and Student-t noise laws are worse by BIC and give 0.88–1.22 when simulated with their own noise (full-sample evaluation, not cross-fitted).

The 2026 postseason. The 33 postseason games are too few to discriminate between no loss and the regular-season loss (Section S6).

11. Discussion.

What a team could do. Within the model, challenge when $p \geq 0.008/(g + 0.008)$ with g in win probability; this is a candidate for prospective testing, not an established prescription. The rule needs the stake in WP, which a team can tabulate before the game by count, inning and score, and an honest confidence. The second is the hard part: the benchmark gives every player a calibrated belief, which nobody has been shown to hold. A practical version is a criterion that falls with the stake, which is what the re-tuned slope of Section 9 does with the information teams have: for a batter under the comparator the one-line rule's signal threshold falls from about 9.1 in in the lowest stake quintile to 0.3 in in the highest, while the teams' criterion moves only from 4.1 to 2.4 in (Table S20). Players cannot be helped by the dugout in the challenge window, so this is a card to learn before the game, not a calculation to make during it. A fixed cost of 0.2 runs, the break-even published on Baseball Savant, performs worst of the rules we evaluate when it is given the model's beliefs; the same form with a tuned cost does well.

Limitations. (i) The engine is league-average and its count transitions are empirical continuations, not interventions on the count; its contrasts agree with Statcast's and with held-out outcomes near the edge, which tests their size on average and in the high-leverage tercile, not every cell. (ii) The counterfactual keeps the real call stream and the opponents' play; the re-simulation shares the engine's structure. (iii) The benchmark is optimal for a pooled opportunity model, and we have no bound on what a richer state could add. (iv) The fielding side pools catchers with 179 pitcher challenges as one decision. The lapse rate is identified by the probit's shape, and its interpretation (attention or choice) is open; the grid over ℓ is the answer we can give. The benchmark's beliefs are learned in cells of the player's information, and finer cells move the index by about 0.1. (v) The interval is conditional on the decision model and the intervention; the model range and the (ρ, ℓ) grid are reported beside it, not folded into it. The held-out worlds are simulated from the training fit, so the cross-fit protects the benchmark's decisions, not the world they are scored in, against the decision model's errors. (vi) Teams' criteria shifted during the season, so the 2026 gap may shrink in 2027, a testable prediction. (vii) The benchmark maximises the probability of winning the game; a player weighing their own plate appearance has a different objective.

Cricket. Cricket's Decision Review System keeps a review that ends in "umpire's call", the same retention structure (Shivakumar, 2018). An exploratory application to Cricsheet data (Rushe, 2026) (Section S7) shows why the known truth matters: without it the decision noise is weakly identified, and the losses, small at the point estimates, are not robust across the noise's own profile interval.

Prior work. The retention DP, the size of the gap and role-specific noise are in public, non-peer-reviewed work that predates this paper (Arora, 2026; Namuduri, 2026; Kirby, 2026; Hires, 2026; Tango, 2026b); Hires (2026) also fits a signal-detection model to all called pitches classified by the zone rule, without a state-dependent criterion or policy evaluation. Descriptive analyses include Lee et al. (2026) and Baumann (2026). The umpire's zone has its own statistical literature (Zimmerman, Tang and Huang, 2019; Deshpande and Wyner, 2017; Mills, 2017). The estimand is a sum of win-probability increments, as in openWAR and nflWAR (Baumer, Jensen and Matthews, 2015; Yurko, Ventura and Horowitz, 2019); the comparison of observed with optimal decisions in sport goes back to Romer (2006), with Sandholtz et al. (2024) the closest template. Off-policy evaluation (Thomas and Brunskill, 2016; Kallus and Uehara, 2020) needs the logging policy's action probabilities given its information, which here depends on a signal we never observe, so policies are valued through the fitted signal model; inference for estimated optimal policies and treatment rules (Hirano and Porter, 2009; Laber et al., 2014; Luedtke and van der Laan, 2016; Andrews, Kitagawa and McCloskey, 2024) motivates the coverage study, and Yam and Lopez (2019) value forgone football decisions with a similar win-probability accounting. Criterion noise and its confounding with perceptual noise are classical in signal detection (Green and Swets, 1966; Benjamin, Diaz and Wee, 2009; Mueller and Weidemann, 2008); what we add is to make that non-identification an explicit dimension of the policy value; the known ABS truth pins down the response curve within the assumed signal model, not the full set of information structures compatible with the decisions.

12. Reproducibility. The code and the outputs are supplied as a supplement to this paper and are available at https://github.com/HeirOfIsildur/abs-challenge-policy. The raw MLB Stats API and Statcast data and the Cricsheet data are public and are not redistributed; the scripts `src/fetch_*.py` re-fetch them, `rerun.sh` rebuilds the MLB results in dependency order from the fetched data (the full policy cross-fit is `src/policy_crossfit.py` and its coverage study `src/coverage_pcf.py`; the bootstraps of the primary model need a cluster, about 915 worker-hours; the cricket pipeline is in `src/drs/`), and `results/manifest.json` records the commit, the package versions and checksums of the raw inputs, since public feeds can be revised. Every number in this paper is written by `paper/make_numbers.py` from `results/`. The exact algebra (Proposition 1, Theorem 2, the intervention identities) is checked in Mathematica 15.0 by the scripts in `wolfram/`; these verify the algebra, not the assumptions under which a result applies to the data (Section S2).

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Declaration of generative AI and AI-assisted technologies in the writing process. During the preparation of this work the authors used large language models (Anthropic's Claude and OpenAI's GPT models) to improve the English of the manuscript, as non-native speakers; to check derivations, formulas and arithmetic; and to assist in writing and reviewing the analysis code. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

SUPPLEMENTARY MATERIAL

Proofs, symbolic verification and additional results

Proofs (S1); the Mathematica checks (S2); the geometry and data audit (S3); the decision models in full (S4); the stakes validation, the policy evaluation, the intervention grid, the causal re-simulation, the bootstraps and the coverage study (S5); the 2026 postseason (S6); the exploratory cricket DRS application (S7).

Code and results

A zip archive of the code repository at the version used for this paper: the data-fetching scripts, the pipeline, the Mathematica scripts in `wolfram/`, the outputs in `results/` from which every number, table and figure is generated, and the manuscript sources. The raw MLB and Cricsheet data are not included; the fetch scripts re-download them from the public sources.

REFERENCES

ABRAMITZKY, R., EINAV, L., KOLKOWITZ, S. and MILL, R. (2012). On the Optimality of Line Call Challenges in Professional Tennis. *International Economic Review* **53** 939–964. https://doi.org/10.1111/j.1468-2354.2012.00706.x

ANDREWS, I., KITAGAWA, T. and MCCLOSKEY, A. (2024). Inference on Winners. *The Quarterly Journal of Economics* **139** 305–358. https://doi.org/10.1093/qje/qjad043

ARORA, A. (2026). Use It or Lose It: The Value and Optimal Use of MLB's 2026 ABS Strike-Zone Challenges. GitHub repository https://github.com/AyanArora29/use-it-or-lose-it. Full-season snapshot, 28 Sep 2026, commit b333b7e; accessed 17 Oct 2026.

ATHEY, S. and WAGER, S. (2021). Policy learning with observational data. *Econometrica* **89** 133–161. https://doi.org/10.3982/ECTA15732

BARLOW, R. E., BARTHOLOMEW, D. J., BREMNER, J. M. and BRUNK, H. D. (1972). *Statistical Inference under Order Restrictions*. London: Wiley.

Baseball Savant. (2026a). ABS Metrics Documentation. https://baseballsavant.mlb.com/abs-metrics-documentation. Breakeven rule $0.2/(0.2 + \mathrm{RV})$ on RE288; accessed 17 Oct 2026.

Baseball Savant. (2026b). Statcast Search CSV Documentation. https://baseballsavant.mlb.com/csv-docs. Entries `plate_x`, `plate_z`; accessed 18 Oct 2026.

BAUMANN, M. (2026). Who's Getting Their Money's Worth From the ABS Challenge System? FanGraphs, 11 Aug 2026, https://blogs.fangraphs.com/whos-getting-their-moneys-worth-from-the-abs-challenge-system/. Accessed 17 Oct 2026.

BAUMER, B. S., JENSEN, S. T. and MATTHEWS, G. J. (2015). openWAR: An open source system for evaluating overall player performance in Major League Baseball. *Journal of Quantitative Analysis in Sports* **11**. https://doi.org/10.1515/jqas-2014-0098

BENJAMIN, A. S., DIAZ, M. and WEE, S. (2009). Signal detection with criterion noise: Applications to recognition memory. *Psychological Review* **116** 84–115. https://doi.org/10.1037/a0014351

BLACKWELL, D. (1953). Equivalent comparisons of experiments. *The Annals of Mathematical Statistics* **24** 265–272.

CHAKRABORTY, B., LABER, E. B. and ZHAO, Y. (2013). Inference for Optimal Dynamic Treatment Regimes Using an Adaptive m-Out-of-n Bootstrap Scheme. *Biometrics* **69** 714–723. https://doi.org/10.1111/biom.12052

CHERNOZHUKOV, V., CHETVERIKOV, D., DEMIRER, M., DUFLO, E., HANSEN, C., NEWEY, W. and ROBINS, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal* **21** 1–68. https://doi.org/10.1111/ectj.12097

CLARKE, S. R. and NORMAN, J. M. (2012). Optimal Challenges in Tennis. *Journal of the Operational Research Society* **63** 1765–1772. https://doi.org/10.1057/jors.2011.147

DESHPANDE, S. K. and WYNER, A. (2017). A hierarchical Bayesian model of pitch framing. *Journal of Quantitative Analysis in Sports* **13** 95–112. https://doi.org/10.1515/jqas-2017-0027

EFRON, B. and TIBSHIRANI, R. J. (1993). *An Introduction to the Bootstrap*. New York: Chapman & Hall.

GOLDSTEIN, J. (2026). Challenge Run Value (cRV): Quantifying Strategic Value in MLB's ABS Challenge Era. Zenodo preprint, 14 Aug 2026. Concept DOI 10.5281/zenodo.21925849; code 10.5281/zenodo.21924168. https://doi.org/10.5281/zenodo.21925850

GREEN, D. M. and SWETS, J. A. (1966). *Signal Detection Theory and Psychophysics*. New York: Wiley.

HAMM, C. (2026). Challenge Leverage Score: A Decision-Support Framework for MLB Automated Ball-Strike System Challenges. Zenodo preprint, version 0.5, 16 Apr 2026. Concept DOI 10.5281/zenodo.19614457. https://doi.org/10.5281/zenodo.19614458

HIRANO, K. and PORTER, J. R. (2009). Asymptotics for Statistical Treatment Rules. *Econometrica* **77** 1683–1701. https://doi.org/10.3982/ECTA6630

HIRES, C. (2026). Perception vs. Decision in MLB's ABS Challenge System. GitHub repository https://github.com/chrishires/abs-sdt. Bayesian signal-detection model; commit 90d2a26, 3 Oct 2026; accessed 17 Oct 2026.

JANOUŠEK, J. and PEŠTA, M. (2026). Supplement to "ABS challenges: wins left on the table". Supplementary Material.

KAKADE, S. and LANGFORD, J. (2002). Approximately Optimal Approximate Reinforcement Learning. In *Proceedings of the Nineteenth International Conference on Machine Learning* 267–274.

KALLUS, N. and UEHARA, M. (2020). Double Reinforcement Learning for Efficient Off-Policy Evaluation in Markov Decision Processes. *Journal of Machine Learning Research* **21** 1–63. https://jmlr.org/papers/v21/19-827.html

KIRBY, K. (2026). The Challenge Economy: What MLB's ABS Challenges Are Actually Worth. GitHub repository https://github.com/kaleb-kirby/abs-challenge-economy. Data through 1 Aug 2026; commit 0c29351; later withdrawn by the author as a pilot; accessed 17 Oct 2026.

KITAGAWA, T. and TETENOV, A. (2018). Who should be treated? Empirical welfare maximization methods for treatment choice. *Econometrica* **86** 591–616. https://doi.org/10.3982/ECTA13288

KLEINBERG, J., LAKKARAJU, H., LESKOVEC, J., LUDWIG, J. and MULLAINATHAN, S. (2018). Human Decisions and Machine Predictions. *The Quarterly Journal of Economics* **133** 237–293. https://doi.org/10.1093/qje/qjx032

LABER, E. B., LIZOTTE, D. J., QIAN, M., PELHAM, W. E. and MURPHY, S. A. (2014). Dynamic treatment regimes: Technical challenges and applications. *Electronic Journal of Statistics* **8** 1225–1272. https://doi.org/10.1214/14-EJS920

LAKKARAJU, H., KLEINBERG, J., LESKOVEC, J., LUDWIG, J. and MULLAINATHAN, S. (2017). The Selective Labels Problem: Evaluating Algorithmic Predictions in the Presence of Unobservables. In *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* 275–284. https://doi.org/10.1145/3097983.3098066

LEE, K., HAN, G., LEE, S. and KO, J. (2026). When the Strike Zone Becomes Algorithmic: Umpire Judgment and Player Challenge Decisions under AI Review. arXiv preprint arXiv:2609.25525 [cs.HC]. Version 1, 22 Sep 2026. https://doi.org/10.48550/arXiv.2609.25525

LUEDTKE, A. R. and VAN DER LAAN, M. J. (2016). Statistical inference for the mean outcome under a possibly non-unique optimal treatment strategy. *The Annals of Statistics* **44** 713–742. https://doi.org/10.1214/15-AOS1384

Major League Baseball. (2025). MLB announces ABS Challenge System coming to the Major Leagues beginning in the 2026 season. https://img.mlbstatic.com/opprops-images/image/upload/opprops/jgdgj1bak2bgiskwpdnm.pdf. Press release, 23 September 2025; accessed 18 Oct 2026.

MANSKI, C. F. (2003). *Partial Identification of Probability Distributions*. New York: Springer.

MILLS, B. M. (2017). Technological Innovations in Monitoring and Evaluation: Evidence of Performance Impacts among Major League Baseball Umpires. *Labour Economics* **46** 189–199. https://doi.org/10.1016/j.labeco.2016.10.004

MUELLER, S. T. and WEIDEMANN, C. T. (2008). Decision Noise: An Explanation for Observed Violations of Signal Detection Theory. *Psychonomic Bulletin & Review* **15** 465–494. https://doi.org/10.3758/PBR.15.3.465

NADIMPALLI, V. K. and HASENBEIN, J. J. (2013). When to Challenge a Call in Tennis: A Markov Decision Process Approach. *Journal of Quantitative Analysis in Sports* **9** 229–238. https://doi.org/10.1515/jqas-2012-0051

NAMUDURI, K. (2026). Who's Leaving Runs on the Table? Optimal ABS Challenge Policy vs. Observed Behaviour. GitHub repository and Streamlit app https://github.com/Cowshal/mlb-abs-challenge-analysis. Commit 66771d7, 16 Sep 2026; accessed 17 Oct 2026.

PRINS, N. (2012). The Psychometric Function: The Lapse Rate Revisited. *Journal of Vision* **12** 25. https://doi.org/10.1167/12.6.25

ROMER, D. (2006). Do firms maximize? Evidence from professional football. *Journal of Political Economy* **114** 340–365. https://doi.org/10.1086/501171

RUSHE, S. (2026). Cricsheet: freely-available structured data for cricket. https://cricsheet.org. Accessed 17 Oct 2026.

SANDHOLTZ, N., WU, L., PUTERMAN, M. and CHAN, T. C. Y. (2024). Learning Risk Preferences in Markov Decision Processes: An Application to the Fourth Down Decision in the National Football League. *The Annals of Applied Statistics* **18** 3205–3228. https://doi.org/10.1214/24-AOAS1933

SHIVAKUMAR, R. (2018). What Technology Says About Decision-Making: Evidence From Cricket's Decision Review System (DRS). *Journal of Sports Economics* **19** 315–331. Published online 11 July 2016. https://doi.org/10.1177/1527002516657218

TANGO, T. (2026a). Cost/Benefit Analysis of Making an ABS Challenge (Part 3 of 3). Tangotiger Blog, 19 Feb 2026, https://tangotiger.com/index.php/site/article/cost-benefit-analysis-of-making-an-abs-challenge. Accessed 17 Oct 2026.

TANGO, T. (2026b). Creating a Naive ABS Challenge Strategy Model. Tangotiger Blog, 26 Jun 2026, https://tangotiger.com/index.php/site/article/creating-a-naive-abs-challenge-strategy-model. Accessed 17 Oct 2026.

THOMAS, P. S. and BRUNSKILL, E. (2016). Data-Efficient Off-Policy Policy Evaluation for Reinforcement Learning. In *Proceedings of the 33rd International Conference on Machine Learning (PMLR)* 2139–2148. https://proceedings.mlr.press/v48/thomasa16.html

WICHMANN, F. A. and HILL, N. J. (2001). The psychometric function: I. Fitting, sampling, and goodness of fit. *Perception & Psychophysics* **63** 1293–1313. https://doi.org/10.3758/BF03194544

YAM, D. R. and LOPEZ, M. J. (2019). What was lost? A causal estimate of fourth down behavior in the National Football League. *Journal of Sports Analytics* **5** 153–167. https://doi.org/10.3233/JSA-190294

YURKO, R., VENTURA, S. and HOROWITZ, M. (2019). nflWAR: A reproducible method for offensive player evaluation in football. *Journal of Quantitative Analysis in Sports* **15** 163–183. https://doi.org/10.1515/jqas-2018-0010

ZIMMERMAN, D. L., TANG, J. and HUANG, R. (2019). Outline Analyses of the Called Strike Zone in Major League Baseball. *The Annals of Applied Statistics* **13** 2416–2451. https://doi.org/10.1214/19-AOAS1285